

A scalar counterexample for nonlinear polynomial Daugavet centers

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Source and question

The source paper is Sheldon Dantas, Miguel Martin, and Yoel Perreau, *The Daugavet property is equivalent to the polynomial Daugavet property*, arXiv:2507.06143 [1].

After proving that every linear Daugavet center is a polynomial Daugavet center, the authors ask on page 11 whether the linearity assumption can be removed:

Given a non-null polynomial $Q \in \mathcal{P}(X, Y)$ satisfying

$$\|Q + T\| = \|Q\| + \|T\|$$

for every rank-one $T \in \mathcal{L}(X, Y)$, must Q be a polynomial Daugavet center, i.e. must the same equality hold when T is a rank-one polynomial from X to Y ?

THE POLYNOMIAL DAUGAVET PROPERTY

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Let us observe that the linearity of the operator G is crucial for adapting the proof of Theorem 2.1 to Daugavet centers (more precisely, to make the ε -quasi-codirected point argument to work), and we do not know if our proof can be adapted in the non-linear setting. In particular, we do not know if, given a (non-null) polynomial $Q \in \mathcal{P}(X, Y)$ which satisfies

$$\|Q + T\| = \|Q\| + \|T\|$$

for every rank-one $T \in \mathcal{L}(X, Y)$, we actually have that Q is a polynomial Daugavet center, that is, the same norm equality holds for T being a rank-one polynomial from X to Y .

3.2 THE WODP and the polynomial WODP are equivalent. In this case

The answer is negative, even in dimension one.

Definitions Used

The source paper defines a polynomial $P : X \rightarrow Y$ as a finite linear combination of homogeneous polynomials, including constant maps as 0-homogeneous polynomials. It defines a rank-one polynomial as a map

$$P(x) = p(x)y_0$$

with $p \in \mathcal{P}(X)$ scalar-valued and $y_0 \in Y$. A non-null polynomial $Q \in \mathcal{P}(X, Y)$ is a polynomial Daugavet center if

$$\|Q + P\| = \|Q\| + \|P\|$$

for every rank-one polynomial $P \in \mathcal{P}(X, Y)$.

Proof Intuition

Linear perturbations of the scalar polynomial $Q(z) = z^2$ are too symmetric to detect the obstruction. If $T(z) = az$, the maximum of $|z^2 + az|$ on the unit ball is attained at a boundary point whose phase aligns z with a , giving exactly $1 + |a|$. Polynomial perturbations are much more flexible: the rank-one polynomial $P = -Q$ cancels Q identically. Thus the linear Daugavet-center test is strictly weaker than the polynomial one.

Theorem 1. *Over either $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$, there are one-dimensional Banach spaces $X = Y = \mathbb{K}$ and a non-null polynomial $Q \in \mathcal{P}(X, Y)$ such that*

$$\|Q + T\| = \|Q\| + \|T\|$$

for every rank-one linear operator $T \in \mathcal{L}(X, Y)$, but Q is not a polynomial Daugavet center.

Proof. Take $X = Y = \mathbb{K}$ with the usual absolute-value norm and define

$$Q(z) = z^2 \quad (z \in \mathbb{K}).$$

Then $\|Q\| = 1$. Every linear operator $T : X \rightarrow Y$ has the form $T(z) = az$ for some $a \in \mathbb{K}$, and in dimension one every nonzero such operator has rank one. The zero operator also causes no issue.

For $|z| \leq 1$, the triangle inequality gives

$$|Q(z) + T(z)| = |z^2 + az| \leq |z|^2 + |a||z| \leq 1 + |a|.$$

Conversely, if $a \neq 0$, choose $z = a/|a|$ in the complex case and $z = \text{sign}(a)$ in the real case. Then $|z| = 1$ and

$$|z^2 + az| = |z||z + a| = 1 + |a|.$$

If $a = 0$, the same equality is clear. Hence

$$\|Q + T\| = 1 + |a| = \|Q\| + \|T\|$$

for every rank-one linear T .

However, $P = -Q$ is a rank-one polynomial from X to Y : indeed

$$P(z) = p(z)y_0, \quad p(z) = z^2, \quad y_0 = -1.$$

It has $\|P\| = 1$, while

$$\|Q + P\| = 0 < 2 = \|Q\| + \|P\|.$$

Therefore Q is not a polynomial Daugavet center. □

Verification Notes

The proof is exact. The accompanying script `code/verify_scalar_example.py` samples the complex unit circle for several values of a and illustrates $\sup_{|z| \leq 1} |z^2 + az| = 1 + |a|$, and records the failure for $P = -Q$.

Novelty Check and Limitations

On 2026-06-20, I searched for exact phrases including

"polynomial Daugavet center" "rank-one" "linear",
 "we do not know" "polynomial Daugavet center",

and the source title. The only direct hit found was the source arXiv paper. This counterexample answers only the nonlinear polynomial-center question; it does not address the separate Bourgain-type lemma question for the weak polynomial topology or the open problem of whether DPr implies WODP.

Human Review Recommendation

This should be reviewed as a likely valid counterexample. The key point is whether the source definition of rank-one polynomial indeed includes $P(x) = p(x)y_0$ for arbitrary scalar polynomial $p \in \mathcal{P}(X)$. Under that definition, $P = -Q$ is rank-one and the counterexample is immediate.

References

- [1] Sheldon Dantas, Miguel Martin, and Yoel Perreau, *The Daugavet property is equivalent to the polynomial Daugavet property*, arXiv:2507.06143, 2025.